

VIGNESH KUMAR E

TECH LEAD | SENIOR FULL-STACK ENGINEER | SOLUTION ARCHITECT

Chennai, Tamil Nadu, India | portfolio: vigneshacme.in | Open to International & Onsite Assignments

PROFESSIONAL SUMMARY

Tech Lead and Full-Stack Architect with 10+ years of enterprise experience engineering distributed, high-performance web systems using **.NET Core, Angular, Node.js, SQL Server, and AWS**. Proven track record in API design, microservices, complex asynchronous processing, and SQL database tuning. Demonstrates hands-on leadership across CI/CD automation, cloud architecture, security, and developer mentoring. Actively integrating **AI Agents, Model Context Protocol (MCP), and Retrieval-Augmented Generation (RAG)** into modern enterprise automation workflows.

TECHNICAL SKILLS & ARCHITECTURE

Languages & Web:	C#, TypeScript, JavaScript, SQL, HTML5, CSS3, JSON
Backend Frameworks:	ASP.NET Core, Web API, EF Core, Node.js, Express.js, WCF, YARP API Gateway
Frontend Frameworks:	Angular, React Native, PWA, RxJS
Cloud & DevOps:	AWS (EC2, ALB, RDS, WAF, S3), Docker, GitLab CI/CD, GitLab Runner, IIS, Linux/Ubuntu, PM2
Databases & Tuning:	SQL Server, MySQL, Oracle Query Store, Execution Plans, Indexing, Concurrency, Locking
Messaging & Distributed:	RabbitMQ, AWS SQS, Redis, SignalR, WebSockets, Async Processing, DLQ
AI & Automation:	AI Agents, MCP, RAG, LangGraph concepts, Qdrant / Vector Databases, Playwright

PROFESSIONAL EXPERIENCE

alfaTKG — Tech Lead

2022 – Present

- **Architectural & Technical Leadership:** Lead cross-functional engineering teams in delivering flagship manufacturing products (JQMS, PTE, AlfaDock) utilizing .NET, Angular, Node.js, and SQL Server.
- **Scalable Async Infrastructure:** Designed a queue-based processing engine using RabbitMQ, AWS SQS, and containerized .NET Worker Services to decouple heavy file/thumbnail workloads from API handlers, enabling independent horizontal scaling.
- **Database & Query Tuning:** Optimized enterprise SQL Server instances by analyzing Query Store, execution plans, locking/blocking, and index strategies to resolve severe CPU bottlenecks under high concurrency.
- **Cloud & CI/CD Engineering:** Architected AWS deployments using EC2, ALB, RDS, and WAF with automated GitLab CI/CD pipelines driving automated deployments to multi-tenant IIS environments.
- **API & System Security:** Implemented YARP API Gateway patterns, JWT authentication, distributed caching via Redis, and rate-limiting to protect mission-critical endpoints.
- **AI Integration R&D:** Driving technical initiatives to integrate AI Agents, RAG architecture, vector databases (Qdrant), and Model Context Protocol (MCP) into core enterprise workflows.

alfaTKG — Software Development Engineer

2015 – 2018

- **Full-Stack Engineering:** Developed business-critical web applications and microservices using C#, .NET Core, SQL Server, and modern JavaScript.
- **Integration & Lifecycle:** Authored modular RESTful APIs, optimized database access queries, and provided tier-3 production incident triage.

Sirpi — Software Engineer

2013 – 2015

- Engineered foundational backend APIs and relational database structures using C# and SQL Server across the full SDLC.

KEY ARCHITECTURE & ENGINEERING HIGHLIGHTS

- **Decoupled Asynchronous Processing:** Replaced legacy sync handlers with RabbitMQ/Redis event buses, reducing API latency for heavy document payloads by eliminating polling patterns via SignalR/WebSockets.
- **High-Availability Cloud Design:** Managed zero-downtime application deployments behind AWS Application Load Balancers with multi-AZ RDS databases.

CLOUD HOSTING & HIT TRACKING GUIDE (FIREBASE ARCHITECTURE)

1. Cloud Hosting: Firebase Hosting (100% Free Tier)

Why Firebase? Provides free SSL, high-speed global CDN routing, custom domain support (`vigneshacme.in`), and direct CI/CD integration with GitHub Actions.

Workflow: Direct push to GitHub `main` branch triggers automated build & deployment to Firebase Hosting.

2. Tracking Resume Hits (Serverless Cloud Resume Challenge Architecture)

Implement a decoupled modern micro-backend to track visits and gain hands-on serverless experience:

- **Frontend Component:** Resume site makes a lightweight `FETCH` request upon load.
- **Firebase Cloud Function:** An HTTP-triggered Node.js / TypeScript API endpoint that runs serverless execution logic.
- **Cloud Firestore / Realtime DB:** Atomically increments visitor counts using `FieldValue.increment(1)`.
- **Google Analytics 4 (GA4):** Paired with Firebase to analyze traffic sources, referral links (LinkedIn vs. resume submissions), session duration, and geographic origins.

3. Hands-On Upgrades to Demonstrate Tech Leadership

- **AI Assistant (RAG / Agentic)** Embed a chatbot on your site powered by an AI Agent & Vector Database (Qdrant/Pinecone) so recruiters can interactively query your experience.
- **Infrastructure as Code (IaC)** Provision your Firebase project, Cloud Functions, and DB security rules using **Terraform** scripts in your repository.
- **Playwright Automation** Build E2E tests to run via GitHub Actions before every deployment to verify UI responsiveness and hit-counter connectivity.
- **Real-Time Observability** Add a live visitor count badge on the frontend powered by Realtime WebSockets/SignalR and monitor errors using Cloud Logging.